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**ELECTRONIC DEVICE THAT RESETS LOG-IN PASSWORD OF
DEFAULT USER, DEVICE MANAGEMENT SYSTEM,
COMPUTER-READABLE, NON-TRANSITORY RECORDING
MEDIUM HAVING DEVICE CONTROL PROGRAM RECORDED
THEREON, AND COMPUTER-READABLE, NON-TRANSITORY
RECORDING MEDIUM HAVING DEVICE MANAGEMENT
PROGRAM RECORDED THEREON**

Abstract

An electronic device includes a communication device, a storage device, and a control device that acts as a device controller. The communication device communicates with an external device. The storage device stores therein a log-in password and a default value of a default user. The device controller resets, upon receipt of a request to reset the log-in password via the communication device, the log-in password of the default user, by rewriting the log-in password to the default value.

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Background/Summary

INCORPORATION BY REFERENCE

[0001] This application claims priority to Japanese Patent Application No. 2024-029526 filed on Feb. 29, 2024, the entire contents of which are incorporated by reference herein.

BACKGROUND

[0002] The present disclosure relates to an electronic device that requires a password to log in, a device management system, a computer-readable, non-transitory recording medium having a device control program recorded thereon, and a computer-readable, non-transitory recording medium having a device management program recorded thereon.

[0003] A network system is widely known that includes multifunction peripherals, an example of the electronic device, and a management server acting as the device management system that manages the multifunction peripherals.

SUMMARY

[0004] The disclosure proposes further improvement of the foregoing technique.

[0005] In an aspect, the disclosure provides an electronic device including a communication device, a storage device, and a control device. The communication device communicates with an external device. The storage device stores therein a log-in password of a default user and a default value. The control device includes a processor, and acts as a device controller, when the processor executes a device control program. The device controller resets, upon receipt of a request to reset the log-in password via the communication device, the log-in password of the default user, by rewriting the log-in password to the default value.

[0006] In another aspect, the disclosure provides a device management system that manages an electronic device. The device management system includes a communication device and a control device. The communication device communicates with an external device. The control device includes a processor, and acts as a device manager, when the processor executes a device management program. The device manager transmits a request to reset a log-in password of a default user, to the electronic device via the communication device.

[0007] In still another aspect, the disclosure provides a computer-readable, non-transitory recording medium having a device control program recorded thereon. The device control program is configured to cause an electronic device including a communication device that communicates with an external device, a storage device that stores therein a log-in password of a default user and a default value, and a processor, to act as a device controller, when the processor executes a device control program. The device controller resets, upon receipt of a request to reset the log-in password via the communication device, the log-in password of the default user, by rewriting the log-in password to the default value.

[0008] In still another aspect, the disclosure provides a computer-readable, non-transitory recording medium having a device management program for managing an electronic device, recorded thereon. The device management program is configured to cause a computer including a communication device that communicates with an external device, and a processor, to act as a device manager, when the processor executes the device management program. The device

manager transmits a request to reset a log-in password of a default user, to the electronic device via the communication device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. **1** is a block diagram showing a configuration of a system according to an embodiment of the disclosure;

[0010] FIG. **2** is a block diagram showing a configuration of a device management system, constituted of a single computer;

[0011] FIG. **3** presents an example of device management information;

[0012] FIG. **4** is a block diagram showing a configuration of an image forming apparatus;

[0013] FIG. **5** presents an example of user management information;

[0014] FIG. **6** presents an example of default password information;

[0015] FIG. **7** presents an example of system connection information;

[0016] FIG. **8** is a block diagram showing an exemplary configuration of a user terminal;

[0017] FIG. **9** is a flowchart showing an operation of an image forming apparatus utilized by a device user;

[0018] FIG. **10** is a flowchart showing an operation of the image forming apparatus, performed when the device user logs in;

[0019] FIG. **11** presents an example of a log-in screen;

[0020] FIG. **12** is a flowchart showing an operation of the image forming apparatus, performed when the password of the default user is changed;

[0021] FIG. **13** presents an example of a password change screen;

[0022] FIG. **14** is a flowchart showing an operation of the device management system, performed upon receipt of a connection request from the image forming apparatus;

[0023] FIG. **15** is a flowchart showing an operation of the image forming apparatus, performed when an instruction to start system connecting action is inputted;

[0024] FIG. **16** presents an example of a system connection screen;

[0025] FIG. **17** is a flowchart showing an operation of the device management system, performed upon receipt of an instruction to reset the password of the default user of the image forming apparatus;

[0026] FIG. **18** and FIG. **19** are flowcharts each showing an operation of the image forming apparatus, performed upon receipt of a password reset request from the device management system; and

[0027] FIG. **20** presents an example of an approval screen.

DETAILED DESCRIPTION

[0028] Hereafter, an embodiment of the disclosure will be described, with reference to the drawings. First, a configuration of a system according to the embodiment of the disclosure will be described. FIG. **1** is a block diagram showing the configuration of the system **10** according to the embodiment.

[0029] As shown in FIG. **1**, the system **10** includes a device management system **20**, for managing an image forming apparatus **30**, exemplifying the electronic device. The device management system **20** may be constituted of a single computer such as a personal computer (PC), or a plurality of computers. The device management system **20** may be configured as a cloud server.

[0030] The device management system **20** is, for example, managed by the manufacturer of the image forming apparatus **30**. The device management system **20** is utilized, for example, by a service person belonging to a service provider that provides the maintenance service for the image forming apparatus **30**.

[0031] The system **10** includes the image forming apparatus **30**, which may be, for example, a multifunction peripheral (MFP) or a single-function printer. The system **10** may include, in addition to the image forming apparatus **30**, one or more image forming apparatuses configured similarly to the image forming apparatus **30**.

[0032] The image forming apparatus **30** is, for example, installed at the location of the office of the user of the image forming apparatus **30**, such as a customer of the service provider. To each of the image forming apparatuses **30**, a device ID is assigned as identification information.

[0033] The system **10** includes a user terminal **40**, for example a PC a smartphone, or a tablet terminal, to be utilized by the user of the device management system **20** such as a service person (hereinafter, “system user”). The system **10** may include, in addition to the user terminal **40**, one or more user terminals configured similarly to the user terminal **40**.

[0034] The device management system **20** and the image forming apparatus **30** can communicate with each other, via the network **11** such as a local area network (LAN) or the internet.

[0035] The device management system **20** and the user terminal **40** can communicate with each other, via the network **12** such as a LAN or the internet.

[0036] FIG. **2** is a block diagram showing a configuration of the device management system **20** constituted of a single computer. As shown in FIG. **2**, the device management system **20** includes an operation device **21**, a display device **22**, a communication device **23**, a storage device **24**, and a control device **25**.

[0037] The operation device **21** includes, for example, a keyboard and a mouse for inputting various operations. The display device **22** includes, for example, a liquid crystal display (LCD) for displaying various types of information. The communication device **23** performs communication with an external device, via a network such as a local area network (LAN) or the internet, or directly through wired or wireless communication.

[0038] The storage device **24** includes a non-volatile memory unit such as a semiconductor memory or a hard disk drive (HDD), for storing various types of information. The control device **25** controls the overall operation of the device management system **20**.

[0039] The storage device **24** can store therein a device management program **24A**, for managing the image forming apparatus **30**. The device management program **24A** may be, for example, installed in the device management system **20** during the manufacturing process thereof, additionally installed in the device management system **20** from an external storage medium such as a universal serial bus (USB) memory, or additionally installed in the device management system **20** from the network.

[0040] The storage device **24** can store therein device management information **24B** for managing the image forming apparatus **30**. FIG. **3** presents an example of the device management information **24B**.

[0041] As shown in FIG. **3**, the device management information **24B**, includes a device ID, serving as the identification information of the image forming apparatus **30**, and an authentication code that the image forming apparatus **30** needs when making connection to the device management system **20**, in association with each of the image forming apparatuses **30**. FIG. **3** shows a part of the content of the device management information **24B**.

[0042] For example, the device management information **24B** shown in FIG. **3** indicates that the image forming apparatus **30** having the device ID “D00001” has the authentication code “123456789”. The authentication code is generated in the device management system **20**.

[0043] As shown in FIG. **2**, the control device **25** includes, for example, a central processing unit (CPU), a read-only memory (ROM) containing programs and various types of data, and a random-access memory (RAM) serving as the operating region for the CPU of the control device **25**. The CPU of the control device **25** executes the program stored in the storage device **24** or the ROM of the control device **25**.

[0044] The control device **25** acts, by executing the device management program **24A**, as a device

manager **25A** that manages the image forming apparatus **30**.

[0045] FIG. **4** is a block diagram showing a configuration of the image forming apparatus **30**. As shown in FIG. **4**, the image forming apparatus **30** includes an operation device **31**, a display device **32**, a printer **33**, a scanner **34**, a communication device **35**, a storage device **36**, and a control device **37**.

[0046] The operation device **31** includes, for example, buttons for inputting various operations. The display device **32** includes, for example, an LCD for displaying various types of information. The printer **33** prints an image on a recording medium such as a paper sheet. The scanner **34** reads an image from a document.

[0047] The communication device **35** performs communication with an external device, via a network such as a LAN or the internet, or directly through wired or wireless communication. The storage device **36** includes a non-volatile memory unit such as a semiconductor memory or an HDD, for storing various types of information. The control device **37** controls the overall operation of the image forming apparatus **30**.

[0048] The storage device **36** can store therein a device management program **36A**, for managing the image forming apparatus **30**. The device management program **36A** may be, for example, installed in the image forming apparatus **30** during the manufacturing process thereof, additionally installed in the image forming apparatus **30** from an external storage medium such as a universal serial bus (USB) memory, or additionally installed in the image forming apparatus **30** from the network.

[0049] The storage device **36** stores therein user management information **36B** for managing the user of the image forming apparatus **30** (hereinafter, “device user”). FIG. **5** presents an example of the user management information **36B**.

[0050] As shown in FIG. **5**, the user management information **36B** includes the user ID serving as the identification information of the device user, the password, and the position with respect to the image forming apparatus **30**, in association with each of the device users. FIG. **5** shows a part of the content of the user management information **36B**.

[0051] In the user management information **36B** shown in FIG. **5**, the user ID “U00001” is the user ID of the default device user of the image forming apparatus **30** (hereinafter, “default user”). The default user is the device manager authorized to perform all types of operation with respect to the image forming apparatus **30**.

[0052] As shown in FIG. **4**, the storage device **36** stores therein default password information **36C**, indicating the default password of the default user (hereinafter, “default password”). In the initial state of the image forming apparatus **30**, the password of the default user in the user management information **36B** accords with the default password indicated by the default password information **36C**.

[0053] FIG. **6** presents an example of the default password information **36C**. As shown in FIG. **6**, the default password may be a simple character string, such as “0000”.

[0054] As shown in FIG. **4**, the storage device **36** can store therein a system connection information **36D**, necessary for the image forming apparatus **30** to establish connection to the device management system **20**. FIG. **7** illustrates an example of the system connection information **36D**.

[0055] As shown in FIG. **7**, the system connection information **36D** includes the uniform resource locator (URL) of the device management system **20**, and the authentication code for establishing connection to the device management system **20**.

[0056] As shown in FIG. **4**, the control device **37** includes, for example, a CPU, a ROM containing programs and various types of data, and a RAM serving as the operation region for the CPU of the control device **37**. The CPU of the control device **37** executes the program stored in the storage device **38** or the ROM of the control device **37**.

[0057] The control device **37** acts, by executing the device control program **36A**, as a device controller **37A** that controls the image forming apparatus **30**.

[0058] FIG. 8 is a block diagram showing an exemplary configuration of the user terminal **40**. As shown in FIG. 8, the user terminal **40** includes an operation device **41**, a display device **42**, a communication device **43**, a storage device **44**, and a control device **45**.

[0059] The operation device **41** includes, for example, a keyboard and a mouse for inputting various operations. The display device **42** includes, for example, an LCD for displaying various types of information. The communication device **43** performs communication with an external device, via a network such as a LAN or the internet, or directly through wired or wireless communication. The storage device **44** includes a non-volatile memory unit such as a semiconductor memory or an HDD, for storing various types of information. The control device **45** controls the overall operation of the user terminal **40**.

[0060] The storage device **44** can store therein a web browser program **44A**. The web browser program **44A** may be, for example, installed in the user terminal **40** during the manufacturing process thereof, additionally installed in the user terminal **40** from an external storage medium such as a USB memory, or additionally installed in the user terminal **40** from the network.

[0061] The control device **45** includes, for example, a CPU, a ROM containing programs and various types of data, and a RAM serving as the operation region for the CPU of the control device **45**. The CPU of the control device **45** executes the program stored in the storage device **44** or the ROM of the control device **45**.

[0062] The control device **45** acts as a web browser **45A**, by executing the web browser program **44A**.

[0063] Hereunder, an operation of the system **10** will be described. First, the operation of the image forming apparatus **30**, utilized by the device user, will be described. FIG. 9 is a flowchart showing the operation of the image forming apparatus **30** utilized by the device user. The device controller **37A** of the image forming apparatus **30** always executes the operation specified in FIG. 9.

[0064] As shown in FIG. 9, the device controller **37A** decides whether the password of the default user in the user management information **36B** accords with the default password indicated by the default password information **36C** (step **S101**).

[0065] Upon deciding that the password of the default user in the user management information **36B** accords with the default password indicated by the default password information **36C** (YES at step **S101**), the device controller **37A** sets the image forming apparatus **30** to such a state that acceptance of operations other than a specific operation is limited (hereinafter, “limited-operation status”) (step **S102**).

[0066] In the limited-operation status, it is permitted, for example, to log in to the image forming apparatus **30** with the user ID of the default user, and to change the password of the default user in the user management information **36B**.

[0067] In the limited-operation status, in contrast, for example, registration of a new device user other than the default user in the image forming apparatus **30**, log-in to the image forming apparatus **30** with the user ID of the device user other than the default user, and input by the device user, of an instruction to start an action for connecting the image forming apparatus **30** to the device management system **20** (hereinafter, “instruction to start system connecting action”), are restricted.

[0068] Upon deciding that the password of the default user in the user management information **36B** does not accord with the default password indicated by the default password information **36C** (NO at step **S101**), the device controller **37A** sets the image forming apparatus **30** to a status different from the limited-operation status (hereinafter, “non-limited-operation status”) (step **S103**).

[0069] After step **S102** or step **S103**, the device controller **37A** returns to step **S101**.

[0070] Hereunder, an operation of the image forming apparatus **30**, performed when the device user logs in, will be described. FIG. 10 is a flowchart showing the operation of the image forming apparatus **30**, performed when the device user logs in.

[0071] The device user can input an instruction to start the action for logging in to the image

forming apparatus **30** (hereinafter, “log-in action start instruction”), through the operation device **31** to the image forming apparatus **30**. When the log-in action start instruction is inputted, the device controller **37A** of the image forming apparatus **30** executes the operation specified in FIG. **10**.

[0072] Referring to FIG. **10**, at step **S121** the device controller **37A** causes the display device **32** to display a screen for logging in to the image forming apparatus **30** (hereinafter, “log-in screen **50**”) (see FIG. **11**).

[0073] FIG. **11** presents an example of the log-in screen **50**. As shown in FIG. **11**, the log-in screen **50** includes a text box **51**, a text box **52**, a log-in button **53**, and a cancel button **54**.

[0074] The text box **51** is a widget for inputting the user ID of the device user. The text box **52** is a widget for inputting the password of the device user. The log-in button **53** is a widget for causing the image forming apparatus **30** to permit the log-in to the image forming apparatus **30**. The cancel button **54** is a widget for cancelling the log-in to the image forming apparatus **30**.

[0075] Referring again to FIG. **10**, after step **S121** the device controller **37A** decides whether the log-in button **53** has been pressed (step **S122**).

[0076] Upon deciding that the log-in button **53** has not been pressed (NO at step **S122**), the device controller **37A** decides whether the cancel button **54** has been pressed (step **S123**).

[0077] Upon deciding that the cancel button **54** has not been pressed (NO at step **S123**), the device controller **37A** returns to step **S122**.

[0078] Upon deciding that the cancel button **54** has been pressed (YES at step **S123**), the device controller **37A** causes the display device **32** to erase the display of the log-in screen **50** (step **S124**), thus finishing the operation specified in FIG. **10**.

[0079] Upon deciding that the log-in button **53** has been pressed (YES at step **S122**), the device controller **37A** causes the display device **32** to erase the display of the log-in screen **50** (step **S125**).

[0080] After step **S125**, the device controller **37A** decides whether the combination of the user ID, inputted in the text box **51** before the log-in button **53** was pressed, and the password inputted in the text box **52** before the log-in button **53** was pressed, is included in the user management information **36B** (step **S126**).

[0081] Upon deciding that the combination of the inputted user ID and the inputted password is not included in the user management information **36B** (NO at step **S126**), the device controller **37A** causes the display device **32** to display a screen indicating the failure in logging in, for a predetermined time (e.g., 10 seconds) (step **S127**), and returns to step **S121**.

[0082] Upon deciding that the combination of the inputted user ID and the inputted password is included in the user management information **36B** (YES at step **S126**), the device controller **37A** permits the log-in with the user ID inputted (step **S128**), and finishes the operation specified in FIG. **10**.

[0083] Hereunder, an operation of the image forming apparatus **30**, performed when the password of the default user is changed, will be described. FIG. **12** is a flowchart showing the operation of the image forming apparatus **30**, performed when the password of the default user is changed.

[0084] The default user can input, in a log-in state to the image forming apparatus **30**, an instruction to start the action for changing the password of the default user (hereinafter, “instruction to start changing default user password”), through the operation device **31** to the image forming apparatus **30**. When such instruction to start changing default user password is inputted, the device controller **37A** of the image forming apparatus **30** executes the operation specified in FIG. **12**.

[0085] Referring to FIG. **12**, at step **S141** the device controller **37A** causes the display device **32** to display a screen for changing the password of the default user (hereinafter, “password change screen **60**”) (see FIG. **13**).

[0086] FIG. **13** presents an example of the password change screen **60**. As shown in FIG. **13**, the password change screen **60** includes a text box **61**, a change button **62**, and a cancel button **63**.

[0087] The text box **61** is a widget for inputting a new password. The change button **62** is a widget

for causing the image forming apparatus **30** to register the new password of the default user. The cancel button **63** is a widget for cancelling the change of the password of the default user.

[0088] Referring again to FIG. **12**, after step **S141** the device controller **37A** decides whether the change button **62** has been pressed (step **S142**).

[0089] Upon deciding that the change button **62** has not been pressed (NO at step **S142**), the device controller **37A** decides whether the cancel button **63** has been pressed (step **S143**).

[0090] Upon deciding that the cancel button **63** has not been pressed (NO at step **S143**), the device controller **37A** returns to step **S142**.

[0091] Upon deciding that the cancel button **63** has been pressed (YES at step **S143**), the device controller **37A** causes the display device **32** to erase the display of the password change screen **60** (step **S144**), thus finishing the operation specified in FIG. **12**.

[0092] Upon deciding that the change button **62** has been pressed (YES at step **S142**), the device controller **37A** causes the display device **32** to erase the display of the password change screen **60** (step **S145**).

[0093] After step **S145**, the device controller **37A** decides whether the password, inputted in the text box **61** before the change button **62** was pressed, accords with the password of the default user in the user management information **36B** (step **S146**).

[0094] Upon deciding that the inputted password accords with the password of the default user in the user management information **36B** (YES at step **S146**), the device controller **37A** finishes the operation specified in FIG. **12**.

[0095] Upon deciding that the inputted password does not accord with the password of the default user in the user management information **36B** (NO at step **S146**), the device controller **37A** decides whether the password of the default user in the user management information **36B** accords with the default password indicated by the default password information **36C** (step **S147**).

[0096] Upon deciding that the password of the default user in the user management information **36B** does not accord with the default password indicated by the default password information **36C** (NO at step **S147**), the device controller **37A** overwrites the password of the default user in the user management information **36B**, with the password inputted in the text box **61** before the change button **62** was pressed (step **S148**), and finishes the operation specified in FIG. **12**.

[0097] Upon deciding that the password of the default user in the user management information **36B** accords with the default password indicated by the default password information **36C** (YES at step **S147**), the device controller **37A** overwrites the password of the default user in the user management information **36B**, with the password inputted in the text box **61** before the change button **62** was pressed (step **S149**).

[0098] After step **S149**, the device controller **37A** decides whether there is a value of the URL and the authentication code, in the system connection information **36D** (step **S150**).

[0099] Upon deciding that there is no value of the URL and the authentication code, in the system connection information **36D** (NO at step **S150**), the device controller **37A** finishes the operation specified in FIG. **12**.

[0100] Upon deciding that there is a value of the URL and the authentication code, in the system connection information **36D** (YES at step **S150**), the device controller **37A** makes access to the URL in the system connection information **36D**, and transmits a request for connection to the device management system **20** (hereinafter, "connection request"), to the device management system **20** (step **S151**). Then, the device controller **37A** finishes the operation specified in FIG. **12**.

[0101] The device controller **37A** includes the device ID of the image forming apparatus **30** and the authentication code in the system connection information **36D**, in the connection request of step **S151**.

[0102] FIG. **14** is a flowchart showing an operation of the device management system **20**, performed upon receipt of the connection request from the image forming apparatus **30**. Upon receipt of the connection request from the image forming apparatus **30**, the device manager **25A** of

the device management system **20** executes the operation specified in FIG. **14**.

[0103] Referring to FIG. **14**, the device manager **25A** decides whether the combination of the device ID and the authentication code, included in the connection request, is included in the device management information **24B** (step **S161**).

[0104] Upon deciding that the combination of the device ID and the authentication code in the connection request is included in the device management information **24B** (YES at step **S161**), the device manager **25A** establishes the connection to the image forming apparatus **30**, which is the transmission source of the connection request (step **S162**).

[0105] After step **S162**, the device manager **25A** notifies that the connection has been established between the device management system **20** and the transmission source of the connection request, in other words the image forming apparatus **30**, to the transmission source of the connection request (step **S163**), and finishes the operation specified in FIG. **14**.

[0106] Upon deciding that the combination of the device ID and the authentication code in the connection request is not included in the device management information **24B** (NO at step **S161**), the device manager **25A** notifies the rejection of the connection between the device management system **20** and the image forming apparatus **30** which is the transmission source of the connection request, to the transmission source of the connection request (step **S164**), and finishes the operation specified in FIG. **14**.

[0107] The system user can confirm, in the log-in state to the device management system **20** realized from the web browser **45A** of the user terminal **40**, via a webpage provided by the device management system **20**, various types of information of the image forming apparatus **30** being managed by the device management system **20**, via the webpage provided by the device management system **20**, from the web browser **45A**.

[0108] The system user can also execute, in the log-in state to the device management system **20** realized from the web browser **45A** of the user terminal **40**, via the webpage provided by the device management system **20**, various operations with respect to the image forming apparatus **30** being managed by the device management system **20**, for example rebooting the same, through the device management system **20**, via the webpage provided by the device management system **20**, from the web browser **45A**.

[0109] Hereunder, an operation of the system **10**, performed when the image forming apparatus **30** is connected to the device management system **20**, in accordance with the instruction to start system connecting action, will be described. FIG. **15** is a flowchart showing the operation of the image forming apparatus **30**, performed when the instruction to start system connecting action is inputted.

[0110] When the image forming apparatus **30** is in the non-limited-operation status, the device user can input the instruction to start system connecting action to the image forming apparatus **30**, through the operation device **31**. When the instruction to start system connecting action is inputted, the device controller **37A** of the image forming apparatus **30** executes the operation specified in FIG. **15**.

[0111] Referring to FIG. **15**, at step **S181** the device controller **37A** causes the display device **32** to display a screen for connecting the image forming apparatus **30** to the device management system **20** (hereinafter, "system connection screen **70**") (see FIG. **16**).

[0112] FIG. **16** presents an example of the system connection screen **70**. As shown in FIG. **16**, the system connection screen **70** includes a text box **71**, a text box **72**, a connect button **73**, and a cancel button **74**.

[0113] text box **71** is a widget for inputting the URL of the device management system **20**. The text box **72** is a widget for inputting the authentication code for making connection to the device management system **20**. The connect button **73** is a widget for causing the image forming apparatus **30** to establish connection to the device management system **20**. The cancel button **74** is a widget for cancelling the connection to the device management system **20**.

[0114] The URL of the device management system **20** and the authentication code are notified in advance from the system user to the device user.

[0115] Referring again to FIG. **15**, after step **S181** the device controller **37A** decides whether the connect button **73** has been pressed (step **S182**).

[0116] Upon deciding that the connect button **73** has not been pressed (NO at step **S182**), the device controller **37A** decides whether the cancel button **74** has been pressed (step **S183**).

[0117] Upon deciding that the cancel button **74** has not been pressed (NO at step **S183**), the device controller **37A** returns to step **S182**.

[0118] Upon deciding that the cancel button **74** has been pressed (YES at step **S183**), the device controller **37A** causes the display device **32** to erase the display of the system connection screen **70** (step **S184**), thus finishing the operation specified in FIG. **15**.

[0119] Upon deciding that the connect button **73** has been pressed (YES at step **S182**), the device controller **37A** causes the display device **32** to erase the display of the system connection screen **70** (step **S185**).

[0120] After step **S185**, the device controller **37A** makes access to the URL inputted in the text box **71** before the connect button **73** was pressed, and transmits the connection request to the device management system **20** (step **S186**).

[0121] The device controller **37A** includes the device ID of the image forming apparatus **30** and the authentication code inputted in the text box **72** before the connect button **73** was pressed, in the connection request of step **S186**.

[0122] Upon receipt of the connection request from the image forming apparatus **30**, the device manager **25A** of the device management system **20** executes the operation specified in FIG. **14**.

[0123] Referring again to FIG. **15**, after step **S186**, the device controller **37A** of the image forming apparatus **30** decides whether the connection between the device management system **20** and the image forming apparatus **30** has been established (step **S187**).

[0124] After step **S186**, for example, the device controller **37A** may decide that the connection between the device management system **20** and the image forming apparatus **30**, in the case where the establishment of the connection between the device management system **20** and the image forming apparatus **30** is notified from the device management system **20**, within a predetermined period of time.

[0125] For example, the device controller **37A** may decide that the connection between the device management system **20** and the image forming apparatus **30** has failed to be established, in the case where the rejection of the connection between the device management system **20** and the image forming apparatus **30** is notified from the device management system **20**, or where the establishment of the connection between the device management system **20** and the image forming apparatus **30** has not been notified from the device management system **20**, within the predetermined period of time after step **S186**.

[0126] Upon deciding that the connection between the device management system **20** and the image forming apparatus **30** has been established (YES at step **S187**), the device controller **37A** overwrites the URL and the authentication code in the system connection information **36D**, with the URL and the authentication code used at step **S186** (step **S188**).

[0127] Upon deciding that the connection between the device management system **20** and the image forming apparatus **30** has not been established (NO at step **S187**), or finishing the operation of step **S188**, the device controller **37A** finishes the operation specified in FIG. **15**.

[0128] Hereunder, an operation of the system **10**, performed when the password of the default user of the image forming apparatus **30** is reset, in accordance with the request from the device management system **20**, will be described.

[0129] There may be cases where the device user of the image forming apparatus **30** wishes to log in to the image forming apparatus **30**, as the default user thereof, for example when the device user of the image forming apparatus **30** wishes to perform an operation with respect thereto, which is

permitted only to authorized users.

[0130] However, normally the device user does not often log in to the image forming apparatus **30** as the default user thereof. Accordingly, the device user of the image forming apparatus **30** may have forgotten the password of the default user of the image forming apparatus **30**, when the device user wishes to log in to the image forming apparatus **30**, as the default user thereof.

[0131] The device user of the image forming apparatus **30** may wish, therefore, to reset the password of the default user of the image forming apparatus **30**, for various reasons including having forgotten the password of the default user. When the device user of the image forming apparatus **30** wishes to reset the password of the default user of the image forming apparatus **30**, the device user can request the system user to reset the password of the default user of the image forming apparatus **30**.

[0132] The system user can instruct, for various reasons including having received the request to reset the password of the default user of the image forming apparatus **30**, from the device user thereof, the device management system **20** to reset the password of the default user of the image forming apparatus **30**, from the web browser **45A** of the user terminal **40** via the webpage provided by the device management system **20**.

[0133] When the connection between the device management system **20** and the image forming apparatus **30** is established, the device manager **25A** of the device management system **20** executes the operation specified in FIG. **17**, upon receipt of the instruction to reset the password of the default user of the image forming apparatus **30**, from the web browser **45A** of the user terminal **40** via the webpage provided by the device management system **20**.

[0134] FIG. **17** is a flowchart showing the operation of the device management system **20**, performed upon receipt of the instruction to reset the password of the default user of the image forming apparatus **30**.

[0135] Referring to FIG. **17**, at step **S201** the device manager **25A** transmits the request to reset the password of the default user of the image forming apparatus **30** (hereinafter, “password reset request”), to the image forming apparatus **30**.

[0136] FIG. **18** is a flowchart showing an operation of the image forming apparatus **30**, performed upon receipt of the password reset request from the device management system **20**. FIG. **19** is a flowchart showing the process that follows the flowchart shown in FIG. **18**.

[0137] Upon receipt of the password reset request from the device management system **20**, the device controller **37A** of the image forming apparatus **30** executes the operation specified in FIG. **18** and FIG. **19**.

[0138] Referring to FIG. **18** and FIG. **19**, at step **S221** the device controller **37A** of the image forming apparatus **30** causes the display device **32** to display a screen for acquiring the approval for resetting the password of the default user (hereinafter, “approval screen **80**”) (see FIG. **20**).

[0139] FIG. **20** presents an example of the approval screen **80**. As shown in FIG. **20**, the approval screen **80** includes a message **81**, an approval button **82**, and a reject button **83**.

[0140] The message **81** is for requesting the device user to approve the resetting of the password of the default user. The approval button **82** is a widget for approving the resetting of the password of the default user. The reject button **83** is a widget for rejecting the resetting of the password of the default user.

[0141] Referring again to FIG. **18** and FIG. **19**, after step **S221** the device controller **37A** sets a timer for notifying the progress of the operation being executed by the image forming apparatus **30** (hereinafter, “notice of operation progress”), to a predetermined value (step **S222**). The predetermined value may be, for example, 10 seconds.

[0142] After step **S222**, the device controller **37A** starts to count down with the timer for notice (step **S223**).

[0143] After step **S223**, the device controller **37A** decides whether the approval button **82** has been pressed (step **S224**).

[0144] Upon deciding that the approval button **82** has not been pressed (NO at step S224), the device controller **37A** decides whether the reject button **83** has been pressed (step S225).

[0145] Upon deciding that the reject button **83** has not been pressed (NO at step S225), the device controller **37A** decides whether the value of the timer for notice has reached 0 seconds or less (step S226).

[0146] Upon deciding that the value of the timer for notice has not reached 0 seconds or less (NO at step S226), the device controller **37A** returns to step S224.

[0147] Upon deciding that the value of the timer for notice has reached 0 seconds or less (YES at step S226), the device controller **37A** transmits the notice of operation progress, indicating that the resetting of the password of the default user is pending for approval, to the device management system **20** (step S227).

[0148] After step S227, the device controller **37A** returns to step S222.

[0149] Upon deciding that the reject button **83** has been pressed (YES at step S225), the device controller **37A** causes the display device **32** to erase the display of the approval screen **80** (step S228).

[0150] After step S228, the device controller **37A** transmits the notice of operation progress, indicating that the resetting of the password of the default user has been rejected, to the device management system **20** (step S229), and finishes the operation specified in FIG. **18** and FIG. **19**.

[0151] Upon deciding that the approval button **82** has been pressed (YES at step S224), the device controller **37A** causes the display device **32** to erase the display of the approval screen **80** (step S230).

[0152] After step S230, the device controller **37A** transmits the notice of operation progress, indicating that the resetting of the password of the default user has been approved, to the device management system **20** (step S231).

[0153] After step S231, the device controller **37A** starts to overwrite the default password indicated by the default password information **36C**, with the password of the default user in the user management information **36B** (step S232).

[0154] After step S232, the device controller **37A** sets the timer for notice to the predetermined value (step S233). The predetermined value may be, for example, 10 seconds.

[0155] After step S233, the device controller **37A** starts to count down with the timer for notice (step S234).

[0156] After step S234, the device controller **37A** decides whether the overwriting of the default password indicated by the default password information **36C**, with the password of the default user in the user management information **36B**, has finished (step S235).

[0157] Upon deciding that the overwriting of the default password has not finished (NO at step S235), the device controller **37A** decides whether the value of the timer for notice has reached 0 seconds or less (step S236).

[0158] Upon deciding that the value of the timer for notice has not reached 0 seconds or less (NO at step S236), the device controller **37A** returns to step S235.

[0159] Upon deciding that the value of the timer for notice has reached 0 seconds or less (YES at step S236), the device controller **37A** transmits the notice of operation progress, indicating that the resetting of the password of the default user is being executed, to the device management system **20** (step S237).

[0160] After step S237, the device controller **37A** returns to step S233.

[0161] Upon deciding that the overwriting of the default password has finished (YES at step S235), the device controller **37A** transmits the notice of operation progress, indicating that the resetting of the password of the default user has been completed, to the device management system **20** (step S238).

[0162] After step S238, the device controller **37A** disconnects the connection between the device management system **20** and the image forming apparatus **30** (step S239), thus finishing the

operation specified in FIG. 18 and FIG. 19.

[0163] Referring again to FIG. 17, after step S201 the device manager 25A of the device management system 20 sets a timer for time-out decision to a predetermined value (step S202). The predetermined value may be, for example, 60 seconds.

[0164] After step S202, the device manager 25A starts to count down with the timer for time-out decision (step S203).

[0165] After step S203, the device manager 25A decides whether the notice of operation progress has been received from the image forming apparatus 30 (step S204).

[0166] Upon deciding that the notice of operation progress has not been received from the image forming apparatus 30 (NO at step S204), the device manager 25A decides whether the value of the timer for time-out decision has reached 0 seconds or less (step S205).

[0167] Upon deciding that the value of the timer for time-out decision has not reached 0 seconds or less (NO at step S205), the device manager 25A returns to step S204.

[0168] Upon deciding that the value of the timer for time-out decision has reached 0 seconds or less (YES at step S205), the device manager 25A displays a message indicating that the time has run out, on the web browser 45A of the user terminal 40, via the webpage provided by the device management system 20 (step S206), and finishes the operation specified in FIG. 17.

[0169] Upon deciding that the notice of operation progress has been received from the image forming apparatus 30 (YES at step S204), the device manager 25A displays the status indicated by the notice of operation progress received from the image forming apparatus 30, on the web browser 45A of the user terminal 40, via the webpage provided by the device management system 20 (step S207).

[0170] After step S207, the device manager 25A decides whether the notice of operation progress received from the image forming apparatus 30 is either indicating that the resetting of the password of the default user has been rejected, or indicating that the resetting of the password of the default user has been completed (step S208).

[0171] Upon deciding that the notice of operation progress received from the image forming apparatus 30 is neither indicating that the resetting of the password of the default user has been rejected, nor indicating that the resetting of the password of the default user has been completed (NO at step S208), the device manager 25A returns to step S202.

[0172] Upon deciding that the notice of operation progress received from the image forming apparatus 30 is either indicating that the resetting of the password of the default user has been rejected, or indicating that the resetting of the password of the default user has been completed (YES at step S208), the device manager 25A finishes the operation specified in FIG. 17.

[0173] Now, in the case of the existing multifunction peripherals, when the user of the multifunction peripheral, who wishes to log in to the multifunction peripheral as the default user thereof, has forgotten the password of the default user for logging in to the multifunction peripheral, the password of the default user has to be reset to a default value. In such a case, it often takes a long time to reset the password of the default user to the default value.

[0174] For example, in the case of having a service person, belonging to a service provider that provides the maintenance work for the multifunction peripheral, visit the site where the multifunction peripheral is installed, and execute the operation to reset the password of the default user to the default value, for the multifunction peripheral in question, or having the user of the multifunction peripheral enquire, for example a customer support center, about how to reset the password of the default user to the default value, and then execute the resetting accordingly with respect to the multifunction peripheral, the time required for resetting the password of the default user to the default value is prolonged.

[0175] According to the foregoing embodiment, in contrast, the image forming apparatus 30 resets, upon receipt of the request to reset the password of the default user, the password of the default user to the default value (i.e., default password) (step S232).

[0176] The mentioned arrangement eliminates the need to have the service person, belonging to the service provider that provides the maintenance work for the multifunction peripheral, visit the site where the multifunction peripheral is installed, and execute the operation to reset the password of the default user to the default value, for the multifunction peripheral in question, or have the user of the multifunction peripheral enquire, for example a customer support center, about how to reset the password of the default user to the default value, and then execute the resetting accordingly with respect to the multifunction peripheral. Therefore, the time required for resetting the password of the default user to the default value can be shortened.

[0177] In addition, upon receipt of the request to reset the password of the default user, the image forming apparatus **30** requests the user of the image forming apparatus **30** to approve the resetting of the password of the default user (step **S221**), and resets, upon acquiring the approval for resetting the password of the default user, from the user of the image forming apparatus **30** (YES at step **S224**), the password of the default user to the default value (step **S232**). Therefore, the password of the default user can be safely reset to the default value.

[0178] Further, when the password of the default user is the default value (YES at step **S101**), the image forming apparatus **30** restricts the operations other than the specific operation from being accepted (step **S102**). Therefore, the security level can be improved.

[0179] The image forming apparatus **30** disconnects the connection to the device management system **20** (step **S239**), upon receipt of the request to reset the password of the default user, from the device management system **20** managing the image forming apparatus **30**, and resetting the password of the default user to the default value (YES at step **S235**).

[0180] Therefore, the image forming apparatus **30** can be prevented from being operated by the device management system **20**, provided that acceptance of the operations other than the specific operation is restricted, because of the password of the default user according with the default value. As result, the security level can be improved.

[0181] When the password of the default user is changed from the default value, the image forming apparatus **30** is automatically reconnected to the device management system **20** (step **S151**). Therefore, the convenience in use can be improved. The electronic device according to the disclosure is exemplified by the image forming apparatus **30**, in the foregoing embodiment. However, the electronic device according to the disclosure may be an electronic device other than the image forming apparatus **30**.

[0182] Further, the computer-readable, non-transitory recording medium, having the device control program according to the embodiment recorded thereon, and the computer-readable, non-transitory recording medium, having the device management program according to the embodiment recorded thereon, can be exemplified by a hard disk, a CD-ROM, a DVD-ROM, or a semiconductor memory.

[0183] While the present disclosure has been described in detail with reference to the embodiments thereof, it would be apparent to those skilled in the art that various changes and modifications may be made therein within the scope defined by the appended claims.

Claims

1. An electronic device comprising: a communication device that communicates with an external device; a storage device that stores therein a log-in password of a default user and a default value; and a control device including a processor, and configured to act, when the processor executes a device control program, as a device controller that resets, upon receipt of a request to reset the log-in password via the communication device, the log-in password of the default user, by rewriting the log-in password to the default value.
2. The electronic device according to claim 1, further comprising: a display device; and an operation device that receives an input of an instruction of a device user, wherein the device

controller causes the display device, upon receipt of the request to reset the log-in password via the communication device, to display an approval screen for acquiring approval for resetting the log-in password, and and resets the log-in password of the default user to the default value, upon receipt of the approval for resetting the log-in password, via the operation device.

3. The electronic device according to claim 2, wherein the device controller restricts acceptance of operations other than a specific operation with respect to the operation device, when the log-in password of the default user accords with the default value.

4. The electronic device according to claim 3, wherein the device controller disconnects connection to a device management system managing the electronic device, upon receipt of the request to reset the log-in password from the device management system via the communication device, and upon resetting the log-in password of the default user to the default value.

5. The electronic device according to claim 4, wherein the device controller reconnects to the device management system, when the log-in password of the default user is changed from the default value.

6. The electronic device according to claim 3, wherein the device controller keeps from restricting the acceptance of the operations other than the specific operation, when the log-in password of the default user discords with the default value.

7. A device management system that manages an electronic device, the device management system comprising: a communication device that communicates with an external device; and a control device including a processor, and configured to act, when the processor executes a device management program, as a device manager that transmits a request to reset a log-in password of a default user, to the electronic device via the communication device.

8. A computer-readable, non-transitory recording medium having a device control program recorded thereon, the device control program being configured to cause an electronic device including a communication device that communicates with an external device, a storage device that stores therein a log-in password of a default user and a default value, and a processor, to act, when the processor executes a device control program, as a device controller that resets, upon receipt of a request to reset the log-in password via the communication device, the log-in password of the default user, by rewriting the log-in password to the default value.

9. A computer-readable, non-transitory recording medium having a device management program for managing an electronic device, recorded thereon, the device management program being configured to cause a computer, including a communication device that communicates with an external device, and a processor, to act, when the processor executes the device management program, as a device manager that transmits a request to reset a log-in password of a default user, to the electronic device via the communication device.
